

Elementary Problems
Extremal Combinatorics
Solutions

Version 1.0

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Basic Work

1. The answer is 2550. If the j th card consists of the numbers $2j - 1$ and $2j$, then the greatest S that we can take is

$$2 + 4 + \cdots + 100 = \frac{(2 + 100) \times 50}{2} = 2550.$$

Next, suppose we always choose the sides with larger numbers. Let these numbers be $a_1, a_2, \dots, a_{50}$ and let the corresponding smaller numbers be $b_1, b_2, \dots, b_{50}$. Then $a_j \geq b_j + 1$ for each j . Thus

$$\begin{aligned} 5050 &= 1 + 2 + \cdots + 100 \\ &= (a_1 + a_2 + \cdots + a_{50}) + (b_1 + b_2 + \cdots + b_{50}) \\ &\leq (a_1 + a_2 + \cdots + a_{50}) + ((a_1 - 1) + (a_2 - 1) + \cdots + (a_{50} - 1)) \\ &= 2(a_1 + a_2 + \cdots + a_{50}) - 50. \end{aligned}$$

This implies $S \geq \frac{5050 + 50}{2} = 2550$. So the maximum k is 2550.

2. The answer is 880. Suppose there is a kind of key which is kept by at most 10 people. When all these people have left, the remaining people do not have the key to open that corresponding locker. Therefore, each kind of key has at least 11 copies. Thus, the total number of keys is at least $11 \times 80 = 880$.

The minimum value 880 can be attained. For example, if there are 11 people each having all the 80 keys, then the condition is clearly satisfied.

Therefore, the minimum value is 880.

3. The answer is 13. For $n \geq 15$, we can easily partition the board into $\left\lfloor \frac{n^2}{2} \right\rfloor$ pairs of adjacent cells, possibly with an additional cell. From the condition, the cells in each pair cannot be both empty. So there is at least 1 coin in each pair. Since

$$\left\lfloor \frac{n^2}{2} \right\rfloor \geq 112 > 111,$$

the condition cannot be satisfied.

For $n = 14$, we colour the cells black and white alternately. Since the numbers of coins in adjacent cells are of different parities, it is obvious that the numbers of coins in the black cells are of the same parity and the same holds for the white cells. Thus, there are $\frac{14^2}{2} = 98$ cells with even numbers of coins and 98 cells with odd numbers of coins. But then the total number of coins is $\equiv 98 \times 0 + 98 \times 1 \equiv 0 \pmod{2}$. This is impossible since there are 111 coins.

The following gives a possibility for the case $n = 13$. There are 85 cells containing 1 coin and 13 cells containing 2 coins. The remaining cells are empty.

1	2	1	2	1	2	1	2	1	2	1	2	1
2	1	2	1	2	1	2	1	2	1	2	1	2
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	0	1	0	1	0	1

4. The cells containing the 2004 smallest numbers in each row cannot be good. So there are at most $n - 2004$ good cells in each row, and hence at most $n(n - 2004)$ good cells. This greatest value can be attained as follows. We put the smallest $2004n$ numbers arbitrarily in the cells $(i, i + j)$ for $i = 1, 2, \dots, n$ and $j = 0, 1, \dots, 2003$ where the entries are taken modulo n . Then there are exactly 2004 such numbers in each row and each column. Therefore, the number in every other cell is larger than 2004 of these numbers in its own row, as well as 2004 of these numbers in its own column. So every other cell is good, yielding a total of $n^2 - 2004n$ good cells.
5. The answer is 6. Indeed, we can cut along two planes parallel to the xy -plane, two planes parallel to the yz -plane, and two planes parallel to the zx -plane in the obvious way to get 27 unit cubes. So 6 cuts suffice.

For the unit cube in the centre, its 6 faces belong to different planes and they cannot be seen initially. For each cut, at most one of these faces can show up. Hence, we need at least 6 cuts to separate this unit cube.

Therefore, the minimum number of cuts needed is 6.

6. The answer is 18. The 9 numbers apart from 1 can be partitioned into 3 consecutive triples of numbers. As their sum is

$$2 + 3 + \dots + 10 = 54,$$

there must be three consecutive numbers with sum $\geq \frac{54}{3} = 18$.

On the other hand, if the numbers are arranged in the order 1, 7, 9, 2, 6, 4, 8, 3, 5, 10 on the circle, then the sum of any three consecutive terms is at most 18. Hence, the maximum value of a is 18.

7. The answer is

- $\left\lceil \frac{n}{3} \right\rceil$ if $n = 3k$ or $n = 3k + 1 \geq 7$;
- 4 if $n = 4$;
- $\left\lceil \frac{n}{3} \right\rceil + 1$ if $n = 3k + 2$.

Since each time we can only get 3 more white cards, at least $\left\lceil \frac{n}{3} \right\rceil$ steps are needed.

When $n = 3k$, we can flip over 3 cards not having been flipped before in each step. Then exactly $\frac{n}{3}$ steps are needed.

When $n = 3k + 1$ with $k \geq 2$, we can first flip over the first $3k - 3$ cards in $k - 1$ steps. Next, we flip over 2 black cards and 1 white card. Lastly, we flip over the remaining 3 black cards. Exactly $k + 1 = \frac{n + 2}{3} = \left\lceil \frac{n}{3} \right\rceil$ steps are needed. When $n = 4$, after 1 step the colours must be WWB where W and B denote white and black cards respectively. After 1 more step, the colours become $WBBW$ (since we should not get back $BBBB$). Clearly, at least 2 more steps are needed. For example, we can flip the cards to get $BWBB$, and then to $WWWW$. At least 4 steps are needed.

When $n = 3k + 2$ with $k \geq 1$, note that we cannot use only $\left\lceil \frac{n}{3} \right\rceil = k + 1$ steps. Indeed, if $k + 1$ steps are used, then since $n < 3(k + 1)$, there must be a card which is flipped more than once. Since each card is flipped for an odd number of times, there is a card which is flipped at least 3 times. But then the number of flipping is at least $n + 2 > 3(k + 1)$. It is possible to use exactly $k + 2$ steps. We can first flip over the first $3k$ cards in k steps. Then we flip over 1 black card and 2 white cards. Lastly, we flip over the remaining 3 black cards.

Therefore, the answer is as claimed.

8. The answer is $\frac{n^2 - 1}{4}$ if n is odd and is $\frac{n^2}{4}$ if n is even.

Let X be the set of x -coordinates used in M and let Y be the set of y -coordinates used in M . If there exists $a \in X \cap Y$, then there are two points $(a, b), (c, a)$ in M , which is a contradiction. Thus, $X \cap Y = \emptyset$.

Conversely, if $X \cap Y = \emptyset$, then we may assume M contains all points (x, y) with $x \in X$ and $y \in Y$ as the condition is satisfied and $|M|$ is maximized. Thus, it remains to maximized $|M| = |X||Y|$ given $|X| + |Y| \leq n$. Clearly, we may assume $|X| + |Y| = n$ and $|X| \leq |Y|$.

If $|Y| - |X| \geq 2$, then we may take one element from Y and put in X . This gives a new product

$$(|X| + 1)(|Y| - 1) = |X||Y| + |Y| - |X| - 1 > |X||Y|.$$

The same process repeats until the difference of $|X|$ and $|Y|$ is at most 1. Thus, when n is odd, the maximum $|M|$ is $\frac{n-1}{2} \cdot \frac{n+1}{2} = \frac{n^2-1}{4}$. When n is even, the maximum M is $\frac{n}{2} \cdot \frac{n}{2} = \frac{n^2}{4}$.

9. The answer is $\frac{n^2}{2}$ if n is even and is $\frac{n^2-1}{2}$ if n is odd.

Let the final positions of $1, 2, \dots, n$ be $f(1), f(2), \dots, f(n)$ respectively. Then the sum is

$$\begin{aligned} S &= |1 - f(1)| + |2 - f(2)| + \dots + |n - f(n)| \\ &= \sum_{k=1}^n (k + f(k) - 2 \min \{k, f(k)\}) \\ &= \sum_{k=1}^n k + \sum_{k=1}^n f(k) - 2 \sum_{k=1}^n \min \{k, f(k)\}. \end{aligned}$$

Since $f(1), f(2), \dots, f(n)$ is a permutation of $1, 2, \dots, n$, they have the same sum. So it is the same as minimizing the sum over $\min \{k, f(k)\}$.

For each integer m , there exist at most two k 's such that $m = \min \{k, f(k)\}$ as f is bijective. Thus, each m appears at most twice in the last sum. It follows that

$$S \leq 2(1 + 2 + \dots + n) - 2(1 + 1 + 2 + 2 + 3 + 3 + \dots).$$

When n is even, the second bracket terminates at two copies of $\frac{n}{2}$. This gives

$$S \leq 2 \cdot \frac{n(n+1)}{2} - 2 \cdot 2 \cdot \frac{\frac{n}{2} \left(\frac{n}{2} + 1 \right)}{2} = \frac{n^2}{2}.$$

When n is odd, the second bracket terminates at a single copy of $\frac{n+1}{2}$. This gives

$$S \leq 2 \cdot \frac{n(n+1)}{2} - 2 \cdot \left(\frac{n+1}{2} \right)^2 = \frac{n^2-1}{2}.$$

The upper bound can be attained when the final configuration is $n, n-1, \dots, 1$. Indeed, in that case we have

$$\begin{aligned} S &= (n-1) + (n-3) + \dots + 3 + 1 + 1 + 3 + \dots + (n-3) + (n-1) \\ &= 2 \cdot \frac{(1 + (n-1)) \cdot \frac{n}{2}}{2} = \frac{n^2}{2} \end{aligned}$$

for even n and

$$\begin{aligned} S &= (n-1) + (n-3) + \cdots + 2 + 0 + 2 + \cdots + (n-3) + (n-1) \\ &= 2 \cdot \frac{(2 + (n-1)) \cdot \frac{n-1}{2}}{2} = \frac{n^2 - 1}{2} \end{aligned}$$

for odd n . Hence, the answer is as claimed.

10. The answer is $2n - 2$. Consider the numbers 1 and n . Suppose $1 = a_1, a_2, \dots, a_k = n$ are the numbers on one of the arcs in that order. By the triangle inequality, we have

$$\begin{aligned} |a_2 - a_1| + |a_3 - a_2| + \cdots + |a_k - a_{k-1}| &\geq |(a_2 - a_1) + (a_3 - a_2) + \cdots + (a_k - a_{k-1})| \\ &= |a_k - a_1| = n - 1. \end{aligned}$$

Similarly, the sum of the differences on the other arc between 1 and n is at least $n - 1$. Then the total sum is at least $2n - 2$. This can be attained when $1, 2, \dots, n$ are written in that order. Hence, the smallest possible value is $2n - 2$.

11. The answer is 8. WLOG assume $1 \in A_j$ for $j = 1, 2, \dots, k$. Then for each pair i, j with $1 \leq i < j \leq k$, the set $A_i \cap A_j$ cannot contain any element other than 1. Thus, the $2k$ elements in one of $A_1, A_2, \dots, A_k$ other than 1 are pairwise distinct. This gives $2k \leq 7$, and hence $k \leq 3$. By symmetry, each element belongs to at most 3 subsets.

Since there are $3m$ elements in the sets $A_1, A_2, \dots, A_m$ counted with multiplicity, and each of the 8 elements appears at most 3, we have $3m \leq 3 \times 8$. This implies $m \leq 8$. Consider the subsets

$$\{1, 2, 3\}, \{1, 4, 5\}, \{1, 6, 7\}, \{2, 5, 8\}, \{3, 6, 8\}, \{4, 7, 8\}, \{2, 4, 6\}, \{3, 5, 7\}.$$

The intersection of any two of them has at most 1 element. So $m = 8$ is possible, and this is the maximum value of m .

12. The answer is

$$\left\lceil \frac{4n}{3} \right\rceil = \begin{cases} 4m & \text{if } n = 3m, \\ 4m + 2 & \text{if } n = 3m + 1, \\ 4m + 3 & \text{if } n = 3m + 2. \end{cases}$$

Let $S = |A_1| + |A_2| + \cdots + |A_n|$. Suppose there are k sets among $A_1, A_2, \dots, A_n$ containing exactly 1 element. Since these sets are all distinct, there are at least k distinct numbers. From the condition, each number appears at least twice. Therefore, we have $S \geq 2k$.

Apart from these k singleton sets, the other $(n - k)$ sets each contains at least 2 elements. Hence, we also have

$$S \geq k + 2(n - k) = 2n - k.$$

Therefore, we obtain

$$3S = S + 2S \geq 2k + 2(2n - k) = 4n.$$

This yields $S \geq \left\lceil \frac{4n}{3} \right\rceil$ since S is an integer.

We show that the lower bound can be attained. For $n = 3m$, we take

$$\{1\}, \{2\}, \dots, \{2m\}, \{1, 2\}, \{3, 4\}, \dots, \{2m-1, 2m\}.$$

For $n = 4$, we take

$$\{1\}, \{2\}, \{3\}, \{1, 2, 3\}.$$

For $n = 3m + 1 \geq 7$, we take

$$\{1\}, \{2\}, \dots, \{2m\}, \{1, 2\}, \{3, 4\}, \dots, \{2m-1, 2m\}, \{1, 2m\}.$$

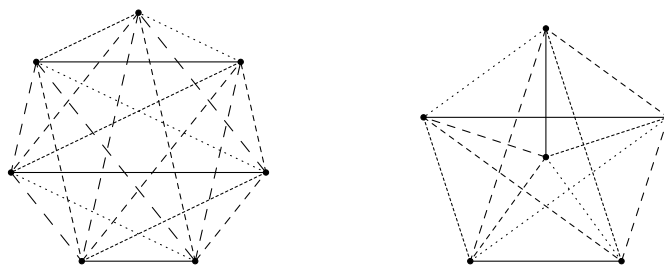
For $n = 3m + 2$, we take

$$\{1\}, \{2\}, \dots, \{2m+1\}, \{1, 2\}, \{3, 4\}, \dots, \{2m-1, 2m\}, \{1, 2m+1\}.$$

Clearly, the condition is satisfied and $S = \left\lceil \frac{4n}{3} \right\rceil$ in these cases.

13. The answer is n . Since each child holds $n - 1$ ribbons and has a hat, at least n colours are needed so that all these things have different colours. It suffices to give a situation for which exactly n colours are used. The case $n = 2$ is trivial.

For odd $n \geq 3$, suppose the children form a regular polygon $P_1 P_2 \dots P_n$. For each side $P_j P_{j+1}$ (where $P_{n+1} = P_1$), there are $\frac{n-3}{2}$ diagonals parallel to this side. We colour this side and these diagonals with the same colour such that any two segments (ribbons) have the same colour iff they are parallel. It follows that no two segments of the same colour share a common vertex. After that, for each vertex, it is the endpoint of $n - 1$ segments of distinct colours. So we can assign a colour to the vertex (hat) which is different from all those segments. This gives a construction for odd n .



For even $n \geq 4$, we first construct the odd case of a regular polygon $P_1 P_2 \dots P_{n-1}$ as above. Suppose the other child is denoted by the centre P_n of the regular polygon. Then we assign the colour to $P_n P_j$ such that it has the same colour as the side of the regular polygon which is perpendicular to $P_n P_j$. Again, the condition is satisfied since there

is no segment passing through P_j which is perpendicular to $P_n P_j$. After that, as only $n - 1$ colours are used, we can simply assign the same new colour to all the vertices. This gives a construction for even $n \geq 4$.

Therefore, at least n colours are needed.

14. The answer is $n + 1$.

Firstly, we claim that the goal can be met if $k = n + 1$. This can be done by the greedy algorithm. For the first paper, we choose the side with the greatest number of integers. So at most $\frac{2^n}{2} = 2^{n-1}$ integers have not appeared yet. Next, for the j th paper where $j = 2, 3, \dots, n$, we always choose the side with the greatest number of integers which have not appeared. So at most one half of those integers remain not appeared. Hence, at most $\frac{2^n}{2^n} = 1$ integer has not appeared after we choose the sides for n pieces of paper. Then we can use the remaining paper to cover this integer.

Secondly, we claim that n pieces of paper is insufficient. For each integer m from 1 to n , we express $m - 1$ in binary representation. We put m on the upper side of the j th paper if the $(j - 1)$ st digit of $m - 1$ in binary representation is 1, and put it on the other side if not. Then each choice of the sides of the paper corresponds to a binary number $\overline{a_0 a_1 \dots a_{n-1(2)}}$ which appears on the side we choose of every paper. Then the binary number $\overline{b_0 b_1 \dots b_{n-1(2)}}$ where $b_j = 1 - a_j$ for each j does not appear on all the paper. So at least $n + 1$ pieces of paper are needed.

15. (a) The answer is 5. Since there are $(n - 1)^2$ copies of 2×2 grid, and there are $2^4 = 16$ possibilities of the colouring, we have $(n - 1)^2 \geq 16$. This yields $n \geq 5$. The case $n = 5$ can be attained. The following gives one possible colouring, where 0 and 1 denote red and blue respectively.

1	1	1	0	1
0	1	0	1	1
1	0	0	1	1
1	0	0	0	0
1	1	0	1	1

- (b) The answer is 10. The construction above shows that 10 red squares are possible. We now show that at least 10 red squares are needed.

If the bottom row is entirely blue, then all the $2^2 = 4$ colourings of 2×2 grids with the second row being blue occur in the 2×2 grids at the two bottom rows. In particular, there are 2 consecutive squares in the fourth row which are both blue. No matter what the colours of the 2 squares above them are, there is another 2×2 grid with the second row being blue. This is a contradiction since there should be no repetitions among the colourings in the 16 copies of 2×2 grids.

Therefore, the bottom row contains exactly 1 red square or at least 2 red squares. In the former case, WLOG assume the bottom left square is blue. Due to symmetry, we

can show that the first column contains at least 1 red square. In any case, the first column and the bottom row together contains at least 2 red squares. If we ignore the first column and the bottom row, then the remaining 4×4 grid should contain 8 more red squares. Indeed, this grid contains all the top right corner squares of the 16 copies of 2×2 grid, and hence half of them are red in colour by symmetry. This gives at least $1 + 1 + 8 = 10$ red squares in total.

Hence, the minimum number of red squares is 10.

Pigeonhole Principle

16. The answer is 68. If all the $100 - 33 = 67$ integers which are not multiples of 3 are chosen, then no 3 of them are consecutive since any 3 consecutive integers must contain a multiple of 3.

We partition the integers as follows:

$$\{1, 2, 3\}, \{4, 5, 6\}, \dots, \{97, 98, 99\}, \{100\}.$$

There are 33 triples and 1 singleton. If 68 numbers are chosen, then there must be $\left\lceil \frac{68-1}{33} \right\rceil = 3$ numbers belonging to the same group, and hence they are consecutive.

Therefore, at least 68 numbers should be chosen.

17. The answer is

$$\begin{cases} 3m+1 & \text{if } n = 4m, \\ 3m+1 & \text{if } n = 4m+1, \\ 3m+2 & \text{if } n = 4m+2, \\ 3m+3 & \text{if } n = 4m+3. \end{cases}$$

Let the vertices be $A_1, A_2, \dots, A_n$.

For $n = 4m$, we group the vertices as follows:

$$(A_1, A_2, A_3, A_4), (A_5, A_6, A_7, A_8), \dots, (A_{4m-3}, A_{4m-2}, A_{4m-1}, A_{4m}).$$

If $3m+1$ vertices are chosen, there must be 4 of them belonging to the same group by the pigeonhole principle. On the other hand, we can choose the $3m$ vertices

$$A_1, A_2, A_3, A_5, A_6, A_7, \dots, A_{4m-3}, A_{4m-2}, A_{4m-1}$$

so that no 4 are consecutive. The minimum value of k is $3m+1$ in this case.

For $n = 4m+1$, WLOG assume A_{4m+1} is not chosen. We use the above partition to show that the condition is satisfied if $k \geq 3m+1$. Also, we can choose the same set of $3m$ vertices so that no 4 are consecutive. The minimum value of k is $3m+1$ in this case.

For $n = 4m + 2$, WLOG assume A_{4m+2} is not chosen. We partition the vertices as follows:

$$(A_1, A_2, A_3, A_4), (A_5, A_6, A_7, A_8), \dots, (A_{4m-3}, A_{4m-2}, A_{4m-1}, A_{4m}), (A_{4m+1}).$$

If $3m + 2$ vertices are chosen, there must be 4 of them belonging to the same group, which are consecutive. If we choose the $3m + 1$ vertices

$$A_1, A_2, A_3, A_5, A_6, A_7, \dots, A_{4m-3}, A_{4m-2}, A_{4m-1}, A_{4m+1},$$

then no 4 are consecutive. The minimum value of k is $3m + 2$ in this case.

For $n = 4m + 3$, WLOG assume A_{4m+3} is not chosen. We partition the vertices as follows:

$$(A_1, A_2, A_3, A_4), (A_5, A_6, A_7, A_8), \dots, (A_{4m-3}, A_{4m-2}, A_{4m-1}, A_{4m}), (A_{4m+1}, A_{4m+2}).$$

If $3m + 3$ vertices are chosen, there must be 4 of them belonging to the same group, which are consecutive. If we choose the $3m + 2$ vertices

$$A_1, A_2, A_3, A_5, A_6, A_7, \dots, A_{4m-3}, A_{4m-2}, A_{4m-1}, A_{4m+1}, A_{4m+2},$$

then no 4 are consecutive. The minimum value of k is $3m + 3$ in this case.

Therefore, the minimum value of k is as claimed.

18. The answer is 2^{n-1} . We can choose 2^{n-1} subsets by choosing all subsets containing 1. Then any two are not disjoint as both contain 1.

On the other hand, we pair up each subset with its complement. There are 2^{n-1} pairs. If we choose at least $2^{n-1} + 1$ subsets, then there are two chosen subsets belonging to the same pair by the pigeonhole principle. But these two subsets are disjoint as they are complement of each other. Thus, we can choose at most 2^{n-1} subsets.

19. The answer is 247. If we erase the 248 integers from 1 to 248, then any 4 remaining integers have sum at least $249 + 250 + 251 + 252 > 1000$. So at most 247 integers can be erased.

Next, consider the following pairs

$$(1, 499), (2, 498), (3, 497), \dots, (249, 251).$$

If 247 integers are erased, then the integers in at least 2 of these pairs are not erased. By choosing the 4 integers in these 2 pairs, their sum is exactly 1000 since each pair has sum 500. Thus, k can be 247.

20. The answer is 52. Firstly, if the 51 integers

$$0, 1, 2, \dots, 50,$$

are chosen, then we have $1 \leq |a - b| \leq 50$ and $1 \leq a + b \leq 99$ for any a and b among these numbers. This shows none of the sums or differences is a multiple of 100.

Now, for any 52 integers, if two of them are congruent modulo 100, then we can simply take these two integers. Thus, we may assume they are pairwise incongruent. We partition the congruence classes as follows:

$$\{0\}, \{1, 99\}, \{2, 98\}, \dots, \{49, 51\}, \{50\}.$$

There are 51 groups in total. By the pigeonhole principle, two chosen integers belong to the same group. Hence, their sum is divisible by 100.

Therefore, the minimum value of k is 52.

21. The answer is $8m + 1$.

Firstly, consider the subset T of S containing those numbers greater than $10m$ and less than $30m$. Note that in the range $10t$ to $10(t + 1)$, there are exactly 4 numbers belonging to S , namely, $10t + 1, 10t + 3, 10t + 7, 10t + 9$. Therefore, T consists of $\frac{4}{10} \times (30m - 10m) = 8m$ numbers. For any two numbers $a < b$ in T , if $a \mid b$, then $3a \leq b$ since they are odd. This gives $b \geq 3a > 30m$, which is a contradiction. Thus, for each $k \leq 8m$, there exists a subset of S not containing two integers with one dividing the other.

It remains to show that $k = 8m + 1$ suffices. Take any element a of S relatively prime to 3 and form the subset $S_a = \{a, 3a, 3^2a, \dots\}$ of S . Note that any element of S belongs to exactly one S_a . There are $\varphi(30)m = 8m$ subsets S_a as $(a, 30) = 1$. By the pigeonhole principle, two of the numbers must belong to the same S_a . Clearly, the larger of them is a multiple of the smaller. This completes the proof.

22. The answer is $n + 1$. We first show that $k \leq n$ is insufficient. Note that 3 vertices of a regular $2n$ -gon form a right-angled triangle if and only if there is a pair of opposite vertices. If $k \leq n$, we can always paint the vertices so that every pair of opposite vertices has the same colour. In that case, we cannot find 3 vertices of a right-angled triangle having pairwise distinct colours.

For $k = n + 1$, there must be a pair of opposite vertices having distinct colours by the pigeonhole principle as there are exactly n pairs. We choose this pair of opposite vertices together with another vertex having a different colour from them. Such a vertex clearly exists since the number of colours is $n + 1 \geq 3$. Then these 3 vertices form a right-angled triangle with pairwise distinct colours.

Therefore, the minimum k is $n + 1$.

23. The answer is $2n - 2$. Firstly, there are $2n - 1$ diagonal lines with positive slopes. Therefore, at most $2n - 1$ cells can be chosen. If exactly $2n - 1$ cells are chosen, then we must choose the cell in the top left corner and the cell in the bottom right corner.

However, these two cells lie in the same diagonal line with negative slope. This is a contradiction. Thus, at most $2n - 2$ cells can be chosen.

Next, we can simply choose all cells in the first column and all cells except the corner cells in the last column. This gives a total of $2n - 2$ cells and the condition is satisfied.

24. The answer is 5. We partition the integers as follows:

$$\{1, 5\}, \{2, 9\}, \{3, 7\}, \{4, 11\}, \{6, 10\}, \{8\}.$$

At most 1 integer can be chosen from each group. If 6 integers are chosen, then we must choose the integer 8.

Symmetrically, we partition the integers as follows:

$$\{7, 11\}, \{3, 10\}, \{5, 9\}, \{1, 8\}, \{2, 6\}, \{4\}.$$

At most 1 integer can be chosen from each group. If 6 integers are chosen, then we must choose the integer 4. This is a contradiction since 4 and 8 have difference 4. Thus, at most 5 integers can be chosen.

One example is to choose 1, 3, 4, 6, 9. Hence, the maximum value is 5.

25. The minimum k is $n + 3$.

Note that no 4 elements in $\{n - 1, n, \dots, 2n\}$ has sum equal to $4n + 1$ since the smallest sum is

$$(n - 1) + n + (n + 1) + (n + 2) = 4n + 2 > 4n + 1.$$

Therefore, $k \geq n + 3$.

Next, we consider any subset S with $n + 3$ elements. Consider the pairs

$$(1, 2n), (2, 2n - 1), \dots, (n, n + 1).$$

Since S contains $n + 3$ elements and there are n pairs, there are at least 3 pairs such that both numbers in each pair belong to S . Let $(a, 2n + 1 - a), (b, 2n + 1 - b), (c, 2n + 1 - c)$ be such pairs where $a, b, c \leq n$.

Similarly, by considering the pairs

$$(1, 2n - 1), (2, 2n - 2), \dots, (n - 1, n + 1),$$

there exists a pair such that both numbers belong to S . Let $(d, 2n - d)$ be such a pair where $d < n$. Since a, b, c are distinct, we may assume $d \neq a, b$. Also, we may assume $2n - d \neq 2n + 1 - a$. Together with $d < n < 2n + 1 - a$ and $2n - d > n \geq a$, the 4 elements $a, 2n + 1 - a, d, 2n - d$ in S are pairwise distinct. Their sum is $4n + 1$ as desired.

So any subset with $n + 3$ elements must contain 4 elements with sum $4n + 1$.

26. The answer is 11.

Firstly, if $m \leq 10$, then there are at most 30 balls. By the pigeonhole principle, one of the boxes contains at most $\left\lfloor \frac{30}{8} \right\rfloor = 3$ balls. For each ball in this box, there are at most 2 other balls with the same number which are put in the other boxes. So at most $2 \times 3 = 6$ other boxes may contain a ball with the same number as this box. This is a contradiction since there are 7 other boxes.

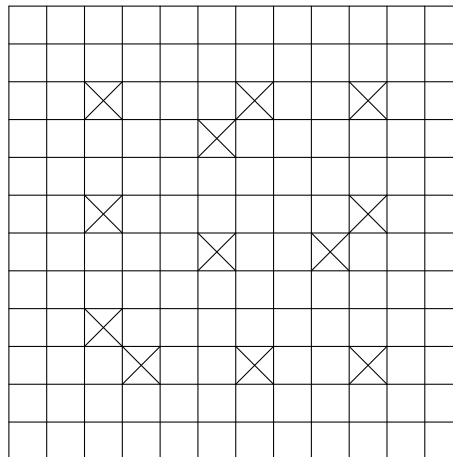
Therefore, we must have $m \geq 11$. The following gives a possible distribution of balls for the case $m = 11$.

$$(1, 2, 3, 4), (1, 5, 6, 7), (1, 8, 9, 10), (2, 5, 8, 11), (2, 3, 6, 9), \\ (3, 7, 10, 11), (4, 5, 9, 10), (4, 6, 7, 8, 11)$$

This shows the minimal m is 11.

27. The answer is 12.

The grid can be decomposed into 12 copies of 3×4 rectangles in the obvious way (with 3 copies in 3 consecutive rows). If there are only 11 black squares, then one black square has to lie in two of the rectangles by the pigeonhole principle. This is a contradiction. So at least 12 black squares are needed. This minimum value can be attained as follows, for example.



28. The answer is 200. Suppose the elements $1, 2, \dots, 1000$ belong to exactly $a_1, a_2, \dots, a_{1000}$ of the 5 subsets $A_1, A_2, \dots, A_5$ respectively. Note that $a_1 + a_2 + \dots + a_{1000}$ is the total number of elements of the 5 subsets, which is $500 \times 5 = 2500$.

We count the total number of elements in the $\binom{5}{2} = 10$ intersections $A_i \cap A_j$. For an element k belonging to exactly a_k subsets, it contributes to $\binom{a_k}{2}$ pairs. Thus,

$$\sum_{i < j} |A_i \cap A_j| = \sum_{k=1}^{1000} \binom{a_k}{2}.$$

By the majorization inequality, the right-hand side is minimized when the variables are closest to each other. Noting that

$$a_1 + a_2 + \cdots + a_{1000} = 2500 = 3 \times 500 + 2 \times 500,$$

this yields

$$\sum_{k=1}^{1000} \binom{a_k}{2} \geq 500 \binom{3}{2} + 500 \binom{2}{2} = 2000.$$

By the pigeonhole principle, at least one of $|A_i \cap A_j|$ is not less than $\frac{2000}{10} = 200$.

To show that $m = 200$ is the greatest possible value, it remains to give a construction that 201 does not work. For simplicity, we use each label $1, 2, \dots, 10$ to denote 100 numbers (thus for example, 1 represents $1, 2, \dots, 100$ while 2 represents $101, 102, \dots, 200$). Consider the 5 subsets

$$\begin{aligned} A_1 &= \{1, 2, 3, 4, 5\}, \\ A_2 &= \{1, 2, 6, 7, 8\}, \\ A_3 &= \{1, 3, 6, 9, 10\}, \\ A_4 &= \{2, 4, 7, 9, 10\}, \\ A_5 &= \{3, 5, 7, 8, 9\}. \end{aligned}$$

Each of them has 500 elements and the intersection of any two of them has 200 elements. This shows $m = 200$ is the best possible.

29. The answer is $\binom{n-1}{m-1}$.

Firstly, if A consists of all r -element subsets of $\{1, 2, \dots, n\}$ containing 1, then

$$|A| = \binom{n-1}{m-1}$$

and the conditions are satisfied.

We claim that this is the maximum value. We consider any permutation $a_1, a_2, \dots, a_n$ of $1, 2, \dots, n$ with $a_1 = 1$. We claim that at most m of the subsets

$$\{a_1, a_2, \dots, a_m\}, \{a_2, a_3, \dots, a_{m+1}\}, \dots, \{a_n, a_1, a_2, \dots, a_{m-1}\} \quad (1)$$

belong to A . WLOG assume $\{a_1, a_2, \dots, a_m\} \in A$. Then only the following subsets have nonempty intersection with A :

$$\begin{aligned} &\{a_{n-m+2}, a_{n-m+3}, \dots, a_n, a_1\}, \{a_2, a_3, \dots, a_{m+1}\}, \\ &\{a_{n-m+3}, a_{n-m+4}, \dots, a_n, a_1, a_2\}, \{a_3, a_4, \dots, a_{m+2}\}, \\ &\dots, \\ &\{a_n, a_1, a_2, \dots, a_{m-1}\}, \{a_m, a_{m+1}, \dots, a_{2m-1}\}. \end{aligned}$$

These $2(m-1)$ subsets can be paired up into $m-1$ pairs of disjoint subsets as shown. This shows at most $m-1$ of them can belong to A . Together with $\{a_1, a_2, \dots, a_m\}$, at most m subsets in (1) belong to A .

There are $(n-1)!$ permutations in total since a_1 is fixed. This gives a total of $m \cdot (n-1)!$ subsets in A counted with multiplicity. Indeed, each subset $\{b_1, b_2, \dots, b_m\}$ is counted exactly $m! \cdot (n-m)!$ times since we can permute these m elements and the $n-m$ elements not in this subset (and then this corresponds to a unique sequence $a_1, a_2, \dots, a_n$ since we have fixed a_1 to be 1). Therefore, we have

$$|A| \leq \frac{m \cdot (n-1)!}{m! \cdot (n-m)!} = \frac{(n-1)!}{(m-1)!(n-m)!} = \binom{n-1}{m-1}.$$

Hence, we conclude that the maximum value of $|A|$ is $\binom{n-1}{m-1}$.

Induction

30. The answer is $2^n - 1$. We first show by induction that $2^n - 1$ steps suffice. The base case $n = 1$ is trivial. Assume this holds for some n . When there are $n+1$ disks, we can first use $2^n - 1$ steps to transfer the top n disks to the second rod. Note that the inductive hypothesis can be applied since the disk with the largest size is always at the bottom and can actually be ignored. Next, we use 1 step to transfer the largest disk to the third rod. Then we use $2^n - 1$ steps to transfer the n disks from the second rod to the third. In total, we use $(2^n - 1) + 1 + (2^n - 1) = 2^{n+1} - 1$ steps. The algorithm works by induction.

Next, we prove by induction that $2^n - 1$ is the minimum value. Again, the base case $n = 1$ is trivial. For the inductive step, since we need to transfer all the disks, the largest disk must be transferred at some stage. In order that it can be transferred, all the other n disks must lie in the same rod other than the first. At least $2^n - 1$ steps are needed before reaching this stage by the inductive hypothesis. Then 1 step is needed to transfer the largest disk. Consider the last moment after the largest disk has been transferred. Again, the other n disks must lie in the same rod. As the largest disk stays

at its position afterwards, we may ignore it and all other disks must be transferred to the corresponding rod. Again, at least $2^n - 1$ steps are needed. This gives at least $2^{n+1} - 1$ steps. By induction, at least $2^n - 1$ steps are required for the case n .

Therefore, the minimum number of steps needed is $2^n - 1$.

31. The answer is $2n - 6$. We first prove that the number of diagonals drawn cannot exceed $2n - 6$ by induction. The case $n = 4$ is trivial since there are only 2 diagonals. Assume all smaller cases hold. Consider the case n . The last diagonal d drawn divides the polygon into two halves, one being a polygon $\mathcal{A}$ with a sides and the other being a polygon $\mathcal{B}$ with b sides. Note that $a + b = n + 2$. In $\mathcal{A}$, at most $2a - 6$ diagonals can be drawn by the inductive hypothesis. Similarly, at most $2b - 6$ diagonals can be drawn in $\mathcal{B}$. The remaining diagonals must join some vertex in $\mathcal{A}$ with some vertex in $\mathcal{B}$, and hence they must intersect d in the interior. So there could be at most 1 such diagonal by the assumption. In total, the number of diagonals drawn is at most

$$(2a - 6) + (2b - 6) + 1 + 1 = 2(a + b) - 10 = 2n - 6.$$

By induction, the upper bound $2n - 6$ holds for any case n .

It remains to give a method for which $2n - 6$ diagonals can be drawn. We again do this by induction. The base case $n = 4$ is trivial since we can draw the 2 diagonals in an arbitrary order. Suppose we can draw $2n - 6$ diagonals in a polygon with n sides. Since $2n - 6 < 2n$, there exists a vertex V of the polygon which is the endpoint of at most 1 diagonal drawn. Consider a polygon with $n + 1$ sides. Let A, B, C, D be 4 consecutive vertices of this polygon. We first draw $2n - 6$ diagonals in the polygon obtained by removing the vertex B (hence AC becomes a side) using the inductive hypothesis, so that C plays the role of V . Then we draw the diagonal BD . This can be done since it only meets those diagonals with C as an endpoint. Afterwards we draw the diagonal AC . This can be done as it only meets the diagonal BD . In total, we have drawn $2(n + 1) - 6$ diagonals.

Therefore, the maximum number of diagonals drawn is $2n - 6$.

32. The answer is n . We first prove by induction that the number of coins that each kid has is divisible by 2^k after the k th interchange where $k \leq n$. The base case $k = 0$ holds trivially.

Consider the inductive step. Suppose the kid with a_0 coins give the coins to the kids with $a_1, a_2, \dots, a_m$ coins respectively in the k th interchange. From the inductive hypothesis, we may assume $2^{k-1} \mid a_j$ for each j . After the k th interchange, the numbers of coins that the kids have become $b, 2a_1, 2a_2, \dots, 2a_m$ respectively. Clearly, we have $2^k \mid 2a_j$ for each $j = 1, 2, \dots, m$. Also, we have

$$b = 2^n - 2a_1 - 2a_2 - \dots - 2a_m,$$

which is divisible by 2^k since $n \geq k$ and each $2a_j$ is divisible by 2^k . So the claim holds by induction. Therefore, after n interchanges (if possible), the number of coins that

each kid has is a multiple of 2^n . This means each kid either has 0 or 2^n coins. Thus, no more interchange is possible.

Next, we show that it is possible to have n interchanges. Suppose there are only 2 kids, one having 1 coin and another having $2^n - 1$ coins. By induction, it is easy to prove that after the k th interchange, one kid has 2^k coins and another has $2^n - 2^k$ coins. We always have $2^n - 2^k \geq 2^{n-1}$ when $k < n$, and so the interchanges can continue until $n = k$.

33. The answer is 2^n . Firstly, for $m = 2^n$, we consider a table whose rows cover all possibilities of $[a_1, a_2, \dots, a_n]$ where $a_j = 0$ or 1 for each j . There are 2^n rows. As any two rows have some entries in the same column which are different, the condition is satisfied.

We now prove by induction that there are at most 2^n rows. When $n = 1$, if there are 3 rows and the numbers in these rows are a, b, c respectively, then the condition implies $|a - b| = |b - c| = |a - c| = 1$. This shows a, b, c have pairwise distinct parities, which is impossible. So there are at most 2 rows. Assume there are at most 2^n rows for the case n . For the case $n + 1$, let a be the smallest entry in the first column. Then all entries in the column must lie between a and $a + 1$.

For those rows (if exist) whose first entries are a , their remaining n columns form a sub-table which satisfies the condition. There are at most 2^n such rows by the inductive hypothesis. For those rows (if exist) whose first entries are not a , the differences of their first entries are less than 1. Hence, their remaining n columns also form a sub-table which satisfies the condition. There are at most 2^n such rows. In total, we get $m \leq 2^n + 2^n = 2^{n+1}$. By induction, the upper bound holds for any n .

Hence, the greatest possible value of m is 2^n .

34. The answer is $m + n$. The case $m + n$ can be attained if we put a rook in each cell of the first column, the first row and also in the bottom right cell.

X	X	X	X
X			
X			
X			X

We prove by induction on $m + n$ that we cannot put at least $m + n + 1$ rooks. The base case is $m + n = 4$ since $m, n \geq 2$. Since there are only 4 cells, we cannot put 5 rooks. For the inductive step, suppose we put at least $m + n + 1$ rooks on an $m \times n$ chessboard. WLOG assume $m \geq n$. As $m + n + 1 \geq 2n + 1$, there exists a column consisting of at least $\left\lceil \frac{2n + 1}{n} \right\rceil = 3$ rooks. Consider a rook in this column which lies between two other rooks in the same column. Since it is threatened by exactly 2 rooks, there is no other rook in the same row as it. By removing this row, the condition remains unchanged.

And we have at least $m + n$ rooks on an $(m - 1) \times n$ chessboard. This contradicts the inductive hypothesis. So we cannot put at least $m + n + 1$ rooks.

By induction, the maximum number of rooks that we can put is $m + n$.

35. The answer is 1999. In general, instead of 1000×1000 , we consider the same problem concerning a $a \times b$ chessboard where $a, b \geq 2$. We claim that the smallest n is $a + b - 1$.

We prove this by induction. If $a = 2$, then we can colour all the cells in the first row without having a coloured right triangle. This shows $n \geq b + 1$. When $n = b + 1$, by the pigeonhole principle, there must be a column containing 2 coloured cells. Since $n \geq 3$, there exists another coloured cell, which forms a right triangle with those 2 cells. By symmetry, the case $b = 2$ is done.

Consider the case $a, b \geq 3$ and assume all the cases (a', b') with $a' \leq a$, $b' \leq b$ and $(a', b') \neq (a, b)$ are true. We can colour every cell in the first row and the first column except the top left corner cell. There are $a + b - 2$ coloured cells and there is no coloured right triangle. Suppose we have coloured $a + b - 1$ cells. WLOG assume $a \geq b$. Then $a + b - 1 \leq 2a - 1$ so that there exists a row with at most 1 coloured cell by the pigeonhole principle.

WLOG assume the first row contains at most 1 coloured cell. If it does not contain any coloured cell, then we simply remove this row and apply the inductive hypothesis to get a coloured right triangle. WLOG we may now assume the top left corner cell is coloured. Suppose there are k other coloured cells in the first column, say from the second to the $(k + 1)$ st row. We are done if one of the second row up to the $(k + 1)$ st row contains another coloured cell. So we may assume each of these rows has exactly 1 coloured cell. We remove the first $k + 1$ rows and the first column. This leaves a $(a - k - 1) \times (b - 1)$ chessboard with $a + b - 1 - (k + 1)$ coloured cells. Note that we must have $a - k - 1 \geq 2$. Indeed, if $a - k - 1 = 1$, then there are $a + b - 1 - (a - 1) = b$ coloured cells in a $1 \times (b - 1)$ chessboard. If $a - k - 1 = 0$, then there are $a + b - 1 - a = b - 1 > 0$ coloured cells in an empty chessboard. Both are impossible. Thus, we can apply the inductive hypothesis to get a coloured right triangle.

By induction, the smallest n for the case $a \times b$ is $a + b - 1$.

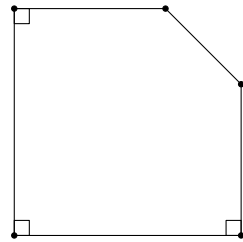
36. The answer is

$$\begin{cases} \frac{2n+3}{3} & \text{if } n \equiv 0 \pmod{3} \text{ and } n \geq 6, \\ \frac{2n+1}{3} & \text{if } n \equiv 1 \pmod{3} \text{ and } n \geq 7, \\ \frac{2n+2}{3} & \text{if } n \equiv 2 \pmod{3} \text{ and } n \geq 8, \\ 3 & \text{if } n = 5. \end{cases}$$

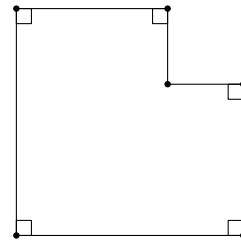
Suppose there are k internal right angles. Each of the remaining $n - k$ angles is at most 360° (note that $n - k > 0$ for $n \geq 5$). Using the sum of internal angles, we have $90k + 360(n - k) > 180(n - 2)$. This gives an upper bound $k < \frac{2n + 4}{3}$. The general case $n \geq 6$ of this bound agrees to the claimed upper bound. It suffices to give a construction for each such case. For $n = 5$, the upper bound only gives $k \leq 4$. However, the case

$k = 4$ is rejected since in that case the remaining angle is 180° , which gives a degenerate case.

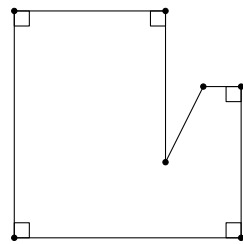
Firstly, we give a construction for each of the base cases $n = 5, 6, 7, 8$.



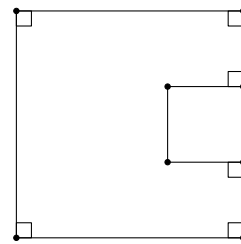
$n = 5$



$n = 6$

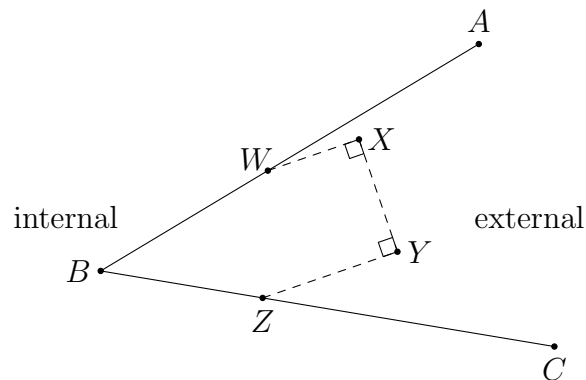


$n = 7$



$n = 8$

Next, we give a construction by induction. Using the base cases, it suffices to show that we can obtain 2 more internal 90° angles by adding 3 more sides to a polygon with at least one reflex internal angle. Consider the following figure which is a reflex internal angle $\angle ABC$ of a given polygon.



The figure itself suggests how to proceed. Choose any point W on AB and any point Z on BC . Choose any point X outside the polygon such that $\angle XWA < \angle CBA$ and the

length WX is sufficiently large. Project X on the line through Z parallel to WX . Let this point be Y . Since the length WX is large, Y lies outside the polygon. Then we use the shape $AWXYZC$ to replace ABC . This increases the number of sides by 3, and we obtain two more internal 90° angles $\angle WXY$ and $\angle XYZ$. Clearly, the internal angle $\angle AWX$ is reflex and the same procedure can be carried out again. This completes the inductive step and hence the proof.

37. The answer is $2n - 1$. Let the letters be $a_1, a_2, \dots, a_n$. The optimal case $2n - 1$ can be attained when the word is $a_1 a_2 \cdots a_n a_{n-1} \cdots a_1$.

We shall prove this upper bound using induction on n . The base case $n = 1$ is obvious. Suppose the maximal length of the word is $2j - 1$ if the alphabet consists of j letters for $j = 1, 2, \dots, k$ where $k \geq 1$. Consider the case with $n = k + 1$ letters. Since we may assume the length is larger than $k + 1$, we may assume a letter a_1 is used more than once. We partition the word into the form

$$\mathcal{I}_{1,1}, a_1, \mathcal{I}_2, a_1, \dots, \mathcal{I}_m, a_1, \mathcal{I}_{1,2}$$

where each $\mathcal{I}_j$ consists of some letters other than a_1 . We combine $\mathcal{I}_{1,1}$ and $\mathcal{I}_{1,2}$ to form a single interval $\mathcal{I}_1$.

If another letter a_2 is used in two different intervals $\mathcal{I}_i$ and $\mathcal{I}_j$, then we can eliminate the extra letters to obtain $a_2 a_1 a_2 a_1$ or $a_1 a_2 a_1 a_2$. This is a contradiction. Therefore, the letters of each interval are different from the letters of other intervals.

Suppose exactly b_j letters are used in the interval $\mathcal{I}_j$ for $j \geq 1$. Then the interval $\mathcal{I}_j$ has length at most $2b_j - 1$ by the inductive hypothesis for $j = 2, 3, \dots, m$. For $j = 1$, it is possible that the last letter of $\mathcal{I}_{1,1}$ is the same as the first letter of $\mathcal{I}_{1,2}$. Thus, the length of $\mathcal{I}_1$ is at most $2b_1$. Together with the m letters a_1 , the total length is at most

$$\begin{aligned} & (2b_1) + (2b_2 - 1) + (2b_3 - 1) + \cdots + (2b_m - 1) + m \\ &= 2(b_1 + b_2 + \cdots + b_m) + 1 \leq 2k + 1 \end{aligned}$$

(the last inequality holds since at most k letters are used in the intervals). This completes the inductive step and the upper bound follows by induction. Thus, the maximal length is $2n - 1$.